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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**
3. Select **Bridged** mode
4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SHH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SHH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngttmpad
    dr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:          0.0
Usage of /:           5.1% of 48.91GB
Memory usage:         5%
Swap usage:           0%
Processes:            139
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

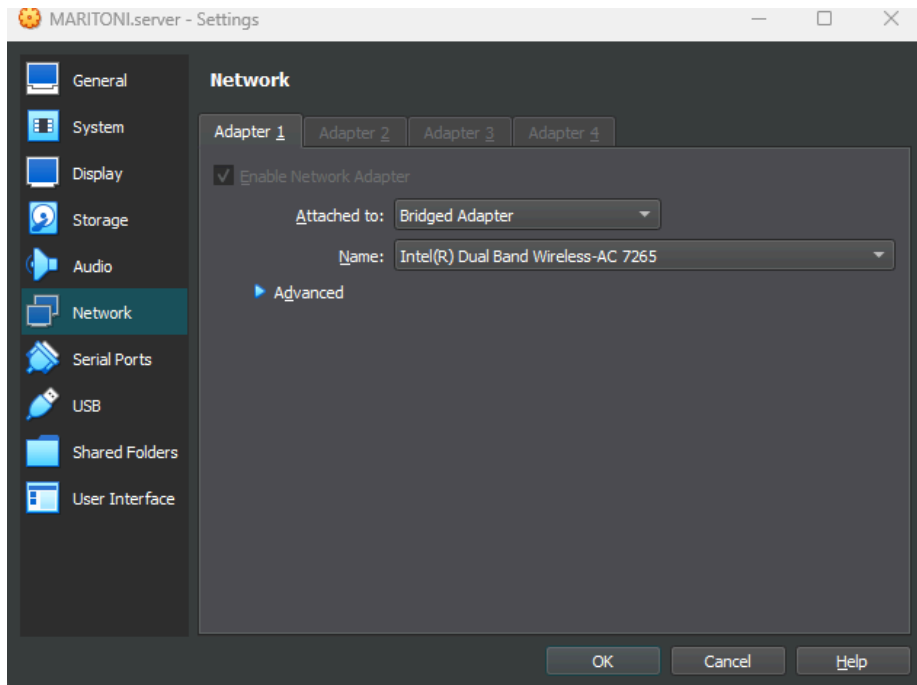
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter
- I changed the VM network setting from NAT into Bridge Adapter.



- Server IP address (output of ip a or hostname -I)
- I got the server IP address after downloading ssh.

```
maritonissn@maritoni:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:00:27:af:a1:f4 brd ff:ff:ff:ff:ff:ff
    altname enx000027afaf4f4
    inet 192.168.137.67/24 metric 100 brd 192.168.137.255 scope global dynamic enp0s3
        valid_lft 604786sec preferred_lft 604786sec
    inet6 fe80::a00:27ff:feaf:a1f4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
maritonissn@maritoni:~$ _
```

- SSH service status (sudo systemctl status ssh)

```
maritonissn@maritoni:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-03 03:31:42 UTC; 30min ago
   Invocation: 38f18a3e3c6045f191d9014784be3ca5
   TriggeredBy: • ssh.socket
                Docs: man:sshd(8)
                  man:sshd_config(5)
   Process: 1304 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1312 (sshd)
   Tasks: 1 (limit: 5974)
   Memory: 6.7M (peak: 8.7M)
   CPU: 167ms
   CGroup: /system.slice/ssh.service
           └─1312 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 03 03:38:22 maritoni sshd-session[1532]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.137.153
Sep 03 03:38:24 maritoni sshd-session[1532]: Failed password for invalid user maritoni from 192.168.137.153 port 56707 ssh2
Sep 03 03:40:19 maritoni sshd[1312]: Timeout before authentication for connection from 192.168.137.153 to 192.168.137.67, pid = 1532
Sep 03 03:41:50 maritoni sshd-session[1540]: Invalid user maritoni from 192.168.137.16 port 35246
Sep 03 03:41:55 maritoni sshd-session[1540]: pam_unix(sshd:auth): check pass; user unknown
Sep 03 03:41:55 maritoni sshd-session[1540]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.137.16
Sep 03 03:41:57 maritoni sshd-session[1540]: Failed password for invalid user maritoni from 192.168.137.16 port 35246 ssh2
Sep 03 03:42:05 maritoni sshd-session[1540]: Connection closed by invalid user maritoni 192.168.137.16 port 35246 [preauth]
Sep 03 03:51:39 maritoni sshd-session[1556]: Accepted password for maritonissn from 192.168.137.16 port 38932 ssh2
Sep 03 03:51:39 maritoni sshd-session[1556]: pam_unix(sshd:session): session opened for user maritonissn(uid=1000) by maritonissn(uid=0)
maritonissn@maritoni:~$
```

- Successful SSH connection from host

```
maritonissn@maritoni:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:af:a1:f4 brd ff:ff:ff:ff:ff:ff
    altname enx080027afa1f4
    inet 192.168.137.67/24 metric 100 brd 192.168.137.255 scope global dynamic enp0s3
        valid_lft 604706sec preferred_lft 604706sec
    inet6 fe80::a00:27ff:feaf:a1f4/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
maritonissn@maritoni:~$ _
```

- Verification commands executed inside VM

```
maritonissn@maritoni: ~ — ssh maritonissn@192.168....
maritoni@maritoni-VirtualBox:~$ ssh maritonissn@192.168.137.67
maritonissn@192.168.137.67's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep  3 03:51:39 AM UTC 2026

System load:  0.0           Processes:            146
Usage of /:   15.1% of 47.93GB Users logged in:          0
Memory usage: 3%           IPv4 address for enp0s3: 192.168.137.67
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

54 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

Name: .

Student ID: 202212154

Course/Year/Section: BSCS 3A

Date Submitted: 09-03-26

Instructor: Mr. Johneros Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☒ Create and configure a Virtual Machine with Ubuntu Server
- ☒ Configure Bridged Network Adapter for network access
- ☒ Determine and verify the server's IP address
- ☒ Install OpenSSH Server and enable the service
- ☒ Establish successful SSH connection from host machine
- ☒ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Ubutu
Virtualization Software (VirtualBox/VMware)	Virtual Box
Ubuntu Server Version	26.04
VM Name	Sison-Electivee3
RAM Allocated	8000 MB
Storage Allocated	100 GB
Network Mode	Bridged Adapter
SSH Client Used	Open SSH

3 VM CONFIGURATION

Setting	Value
VM Name	MARITONI.server
OS Type	Ubuntu (64-bit)
Memory	8000 MB
Hard Disk	100 GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed:

- sudo systemctl status ssh
 sudo apt update
 sudo apt install openssh-server -y
 sudo systemctl enable ssh
 sudo systemctl start ssh
- ip a
- ssh maritonissn@192.168.137.67

Server IP Address: 192.168.137.67

Network Interface: enp0s3

6. REMOTE CONNECTION

Connection Details: enp0s3

Parameter	Value
Server IP	192.168.137.67
Username	maritonissn
SSH Port	Port 22